Market Scale, Political Size, and Canada's Future

AI-enhanced project handbook — choose one option

Work in pairs • Exactly four narrative pages • One week • Auditable evidence

Names _____ **Section** _____

The project. Write a short decision document that lets a busy reader understand a difficult Canadian tradeoff and act on it. This is our **four-page adaptation of an Amazon-style narrative memo**: no slide deck, no collection of bullet points, and no long literature review. Lead with an answer, explain the economic mechanism in ordinary language, test it with evidence, confront the strongest objection, and finish with a conditional decision.

1. Choose one decision

	Option A: The Size of Nations	Option B: 100 Million Canadians
Decision	Which scale benefits can Canada obtain through economic integration while retaining political autonomy?	Under what spatial and policy conditions could 100 million residents raise Canadian living standards?
Core tension	A larger market and shared public-good costs versus policy fit, distribution, and autonomy.	Market scale and urban agglomeration versus housing, congestion, infrastructure, and adjustment costs.
Required result	Recommend one institutional arrangement under stated conditions.	Recommend a spatial growth path under measurable capacity conditions.

2. Submit one package per pair

1. One **exactly four-page main memo** in PDF, including title, byline, footnotes, and exhibits. Use 11-point type, ordinary margins, readable labels, and no separate cover page.
2. A **references and data appendix** of at most one page.
3. A **reproducibility and AI-use appendix** of at most one page, including the model/tool used, important prompts, retained output, checks, corrections, and each partner's contribution.
4. The **data and code** needed to reproduce every reported number and exhibit. Upload the PDF and a ZIP file containing the research folder.

The appendix is for verification, not overflow. Necessary arguments, figures, qualifications, and the recommendation belong in the four-page memo. Material found only in an appendix receives no substantive credit. Canvas controls the exact deadline and submission record.

3. The common economic lens

If producing a variety requires fixed cost (F), then $AC(q) = F/q + c$. A larger market may support more output per variety, lower average cost, more viable varieties, or some combination. That potential is not a verdict: integration, geography, competition, capacity, and political institutions determine whether scale improves welfare.

The Four-Page Narrative

Each page has one job; together they produce a decision-ready argument

4. Exact page architecture

Page	Job	What the reader must learn
1	Decision and answer	Name the decision-maker, baseline, and counterfactual. The first substantive paragraph gives your answer and the two or three forces behind it.
2	Economic mechanism	Explain the causal chain and the countervailing force in ordinary language. Define every technical term. Use one compact conceptual exhibit only if it makes a claim.
3	Evidence and benchmarks	Present two principal exhibits: one baseline and one comparison or counterfactual. Explain what each establishes, what it cannot establish, and why it changes the decision.
4	Tradeoffs and verdict	Address the strongest objection, identify who gains and bears costs, state the conditions for your recommendation, name the main uncertainty, and say what observation would change your mind.

An answer-first opening. "Canada can obtain most North American scale benefits without political union, provided that . . . " or "A Canada of 100 million could raise living standards only if . . . " is useful. "This memo will discuss the advantages and disadvantages of . . . " is not an answer.

5. Narrative-writing standard

The main memo uses complete paragraphs. Do not use slide fragments, decorative graphics, oversized quotations, or a list of generic pros and cons.

- Put the conclusion first; begin each paragraph with its main claim.
- Use claim-based headings, such as "Housing capacity is the binding constraint," rather than generic headings such as "Background."
- Distinguish observed facts, model implications, inferences, scenarios, and value judgments.
- Interpret every exhibit in prose. Do not make the reader reverse-engineer your argument from a chart.
- Explain uncertainty without hiding behind it. Finish with a decision, not merely "more research is needed."

6. Exhibit standard

Use **two principal exhibits**; a third is allowed only if the document remains legible. Recreate figures from downloaded data rather than pasting website screenshots. Every exhibit needs:

- a title that states its substantive conclusion;
- labelled axes, units, geography, and years;
- a source and note explaining transformations and assumptions; and
- a paragraph explaining relevance, limitations, and implications.

Minimum analytical bar. A reader should be able to identify your decision, recommendation, mechanism, two benchmarks, strongest counterargument, sensitivity check, and reversal condition without opening the appendix.

Option A — The Size of Nations

A neutral institutional-design counterfactual, not advocacy about identity or sovereignty

7. Context and goal

Alesina and Spolaore organize political borders around two opposing forces. A larger tax base can spread the fixed cost of non-rival public goods such as defence, diplomacy, regulation, and national institutions. Yet a single common policy may fit citizens less well when preferred taxes, regulation, redistribution, or public services differ.

Trade changes this tradeoff. A politically small country may give its firms access to a large market without pooling all political decisions. Thus, **the scale of the market need not equal the size of the country**. This connects directly to the Week 6 Krugman model.

Your decision question. Advise a hypothetical Canadian institutional-design commission: should Canada retain the status quo, seek deeper economic integration while remaining sovereign, or favour full Canada–U.S. political integration? Recommend one arrangement under explicit conditions. Treat political heterogeneity as differences in preferred public policy, never as a claim that cultural or demographic diversity is a loss.

8. Compare the same three arrangements

Arrangement	Potential scale mechanism	Central cost or limitation
Status quo: sovereign Canada + CUSMA	Large private market with separate national government	Remaining border frictions, rules of origin, bargaining, and policy spillovers
Deeper integration without union	More mutual recognition, mobility, or coordinated infrastructure	More shared rules may constrain autonomy; cooperation remains incomplete
Hypothetical political integration	Common market plus pooled public institutions and fiscal capacity	Policy mismatch, representation, redistribution, transition, and loss of autonomous choice

9. Evidence benchmarks

A successful memo must:

- distinguish population, GDP, trade exposure, and accessible market;
- quantify Canada's existing economic relationship with the United States;
- identify one benefit achievable through CUSMA and one that plausibly requires common political institutions;
- use a direct policy-preference measure, or clearly label institutional outcomes as imperfect proxies; and
- compare all three alternatives on the same criteria: private-market access, public-good scale, policy fit, fiscal risk-sharing, distribution, sovereignty, and transition cost.

Required sensitivity logic. State whether your recommendation survives (1) a different reasonable comparison group and (2) a larger assumed residual cost of crossing the Canada–U.S. border. If it reverses, explain why.

Option A — Evidence Plan

Choose a small set of measures that can distinguish the institutional alternatives

10. Strong exhibits

Use no more than two figures and one compact comparison table.

- **Country size and openness:** trade as a share of GDP against log population for OECD countries, highlighting Canada and pre-specified peers. Treat the fitted relationship as correlation, not causation.
- **Existing integration:** U.S. shares of Canadian merchandise exports and imports over time. Never relabel goods trade as goods and services.
- **Policy fit:** comparable Canada–U.S. survey responses on two or three pre-registered policy questions, showing samples and uncertainty.
- **Scenario table:** apply the same criteria to all three arrangements. Do not disguise unsupported judgments as a composite index.

11. Data and source starter pack

Purpose	Primary or original source
Theory	Alesina and Spolaore, "On the Number and Size of Nations"; Alesina, Spolaore, and Wacziarg, "Economic Integration and Political Disintegration"
Population, GDP, openness	World Bank Indicators API: `SP.POP.TOTL`, `NY.GDP.MKTP.CD`, `NY.GDP.PCAP.PP.KD`, `NE.TRD.GNFS.ZS`
Canadian bilateral trade	Statistics Canada Table 12-10-0171-01, merchandise trade by country and product section
CUSMA institutions	Global Affairs Canada economic impact assessment and official agreement material
Optional policy/public goods	World Values Survey; OECD government spending; SIPRI military expenditure

12. Verification checks

- Use the same year or disclose mismatches; keep currencies, prices, and denominators consistent.
- Do not mix export, import, and total-trade shares; do not call trade-to-GDP literal market size.
- Identify the EU-27 as a bloc, not another country in a country regression.
- Save plotted observations separately; report missing data and exclusions.
- Rerun the size–openness result without microstates and unusual resource exporters. State whether the conclusion changes.
- Never equate a demographic characteristic with a policy preference.

13. Prompts that advance the analysis

"Convert our decision into testable claims, required evidence, and observations that could overturn each claim. Do not answer the question yet."
"Audit this figure's denominator, units, year, exclusions, and caption. Flag causal language the data cannot support."
"Read the memo as an opposing policy analyst. Identify the three weakest links between evidence and recommendation and propose one feasible sensitivity check for each."

Option B — 100 Million Canadians

A spatial capacity scenario, not a forecast and not a referendum on immigration

14. Context and goal

A larger population can create **national market scale**: more buyers may support more varieties and fixed-cost activities. But Canada already trades, so you must explain which benefits remain after firms can sell abroad.

It can also create **urban agglomeration**: nearby firms and workers can share specialized inputs and infrastructure, match in thicker labour markets, and learn through knowledge spillovers. Population scattered across a vast territory does not automatically create those gains. Housing scarcity, congestion, infrastructure pressure, environmental costs, and unequal access to productive cities can offset them.

Your decision question. Advise the Government of Canada whether a population of 100 million by 2100 should be pursued as an economic strategy and, if so, what urban system and capacity plan would make it welfare-improving. Say where growth occurs, what must be built, and which measurable conditions must hold. The test is living standards per person, not mechanically larger GDP.

15. Compare spatial counterfactuals

At minimum compare:

1. **Concentrated growth:** most additional residents enter today's largest metropolitan areas and corridors; and
2. **Polycentric growth:** more growth enters mid-sized metropolitan areas and expanded or emerging urban corridors.

Each scenario must sum to 100 million nationally. You may recommend one, reject both, or propose a hybrid, but must attach three to five measurable guardrails.

16. Numerical anchors

Statistics Canada's 2026 projections begin at about **41.7 million in 2025**. By 2075 the range is **44.0 to 75.8 million**; medium scenario M1 is **57.4 million**. These are scenarios, not predictions. A constant path from 41.7 million in 2025 to 100 million in 2100 requires approximately

$$g = \left(\frac{100}{41.7}\right)^{1/75} - 1 = 1.17\% \text{ per year.}$$

It reaches about 74.7 million in 2075, a useful arithmetic cross-check against the official high scenario—not a claim that the official path continues to 2100.

Housing is a near-term capacity benchmark. CMHC estimates roughly 430,000–480,000 annual starts would be needed through 2035 to restore 2019 affordability, versus projected starts of roughly 245,000–250,000. Do not mechanically extend that estimate to 2100.

Option B — Evidence Plan

The location and capacity of growth determine whether scale becomes welfare

17. Strong exhibits

Require two; use at most three.

- **Population paths:** observed/base population, Statistics Canada M1 and high scenarios through 2075, and your explicit path to 100 million in 2100.
- **Spatial allocation:** a map or table showing where the extra population goes under concentrated and polycentric scenarios. Totals must add.
- **Scale–cost dashboard:** compare scenarios using GDP or productivity per person and one urban-cost measure: shelter burden, homes per added household, commuting, transit access, or infrastructure per resident.

18. Core Canadian data bundle

Choose two to four datasets; do not collect everything.

Purpose	Official source
National projections	Statistics Canada Tables 17-10-0057-01 and 17-10-0058-01
Metropolitan population	Tables 17-10-0148-01 and 17-10-0149-01
CMA GDP and productivity proxy	Table 36-10-0468-01; disclose nominal/real limits and align population year
Density, households, commuting	2021 Census Profile; commuting Table 98-10-0460-01
Housing affordability	Table 98-10-0253-01; CMHC starts and completions
Infrastructure	Canada's National Infrastructure Assessment

19. Verification checks

- **Scenario identity:** regional totals equal 100 million; the path ends at 100 million in 2100.
- **Stock–flow identity:** distinguish people, households, dwellings, starts, completions, and the housing stock. Show conversions.
- **Per-person test:** report productivity or GDP per person plus an affordability or distribution measure; total GDP is not enough.
- **Geography/year/price test:** do not silently mix municipality, CMA, or province; disclose year mismatches and current versus real dollars.
- **Causality test:** city-size correlations are not automatically causal because firms and workers select into cities.
- **Capacity and sensitivity:** compare required building rates with recent rates; vary one central assumption and report whether the verdict changes.

Option B — Decision Benchmarks

Translate a long-run population total into measurable conditions for living standards

20. What success must mean

Your recommendation must discuss all four objects separately:

Object	Useful measure	Common mistake
Economic output	Total real GDP	Calling a mechanical increase in aggregate GDP a welfare gain
Living standards	Real GDP or income per person	Ignoring housing and local price differences
Productivity	Output per hour or a carefully labelled proxy	Treating city size as a causal productivity treatment
Urban welfare	Shelter burden, access, commuting, services, environment	Reporting a national average while constraints are local

21. Recommended guardrails

Choose three to five and justify the thresholds. Examples include:

- dwellings completed per additional household;
- no increase in the share spending 30% or more of income on shelter;
- commute-time or frequent-transit-access targets;
- water, energy, school, health, and transport capacity delivered before or alongside occupancy;
- a minimum improvement in real GDP per person or labour productivity; and
- a regional distribution check showing which households gain and pay.

22. Prompts that advance the analysis

"Build a source inventory using only the official tables in the handbook. Record table ID, URL, access date, years, geography, units, and limitations. Do not calculate anything yet."
"Construct concentrated and polycentric allocations. Assert that each sums to exactly 100 million. Label every observed value, assumption, and extrapolation."
"Audit this analysis for total-versus-per-capita errors, stocks versus flows, geography/year mismatches, nominal-versus-real confusion, correlation presented as causation, unsupported forecasts, and arithmetic that does not add up."

23. Original urban-economics context

Start with Duranton and Puga's sharing–matching–learning framework and their review of urban costs and benefits. For Canadian evidence, consult Statistics Canada's review of agglomeration economies and productivity. Use the Century Initiative only as a labelled advocacy source; it is not the official forecast.

Interpretation rule. This topic is about the economic consequences of a population scenario and the institutions required to manage it. It is not a referendum on immigration or a claim that population is an end in itself.

Codex Research Workflow

Use AI to accelerate auditable inquiry, not to outsource judgment

24. Create a safe project folder

Work only in a dedicated folder that contains course data—never in a broad home, Dropbox, or system folder. A useful structure is:

```
project/
  README.md            sources.md           AI_LOG.md
  data/raw/            data/processed/      code/
  figures/             memo/                verification_report.md
```

Preserve downloaded files unchanged in `data/raw`. Generated data belong in `data/processed`. Your README should explain how to reproduce the analysis from a clean start.

25. A one-week sequence

Day	Objective	Research-assistant workflow
1	Work backward	Name the decision-maker, baseline, counterfactual, tentative answer, mechanisms, needed evidence, and possible refutations.
2	Acquire and audit	Download primary-source files; record metadata, definitions, units, geography, years, and access dates. Never overwrite raw data.
3	Build benchmarks	Write reproducible code; assert units, missingness, duplicates, totals, and scenario endpoints. Manually check three observations.
4	Draft the argument	Write the answer paragraph first, then mechanism, evidence, objection, and conditional recommendation.
5	Red-team	Classify claims as fact, model implication, inference, scenario, or recommendation; identify unsupported or causal overclaims.
6	Reproduce and revise	Run from raw data in a clean folder; regenerate figures and fix the logic and prose.
7	Final audit	Read as the decision-maker; test links, exhibit legibility, page count, files, and contribution statement.

26. High-value prompts

Plan: "Do not answer the policy question. Act as a skeptical research supervisor. Identify ambiguities, missing causal links, and evidence that would distinguish competing explanations."
Data: "Inspect the files and documentation before coding. Produce a data dictionary using actual column names. Flag incompatible units, years, geographies, and missing values. Do not invent variables."
Figure: "Write reproducible code using only documented variables. Add assertions for units, duplicates, missing values, and totals. Use a claim-based title and source note."
Red team: "Audit the memo sentence by sentence. Flag unsupported claims, causal overstatements, inconsistent benchmarks, and places where an exhibit does not support the prose. Do not rewrite it yet."

Never put passwords, tokens, private student data, or confidential information in a prompt. AI output is not a source. You remain responsible for every number, citation, inference, and sentence you submit.

Verification, Rubric, and Submission

The bar is a concise document that another researcher can audit

27. Universal verification gate

Both partners must be able to answer *yes*:

- Can every number be traced to a named dataset, variable, and raw row?
- Are years, currencies, price bases, units, geographies, and denominators consistent or clearly disclosed?
- Did we manually verify at least three observations and compare an aggregate with an official published value where possible?
- Did we inspect missing values and exclusions and keep raw data unchanged?
- Do all exhibits regenerate from a clean run of the submitted code?
- Does each factual claim have a source, and does each link work in the PDF?
- Did we perform the required sensitivity check and state whether the recommendation changes?
- Did we document important AI uses, retained output, errors, and corrections?

28. Assessment benchmarks — 100 points

Dimension	Weight	Excellent work
Decision question and answer-first opening	15	The decision, baseline, counterfactual, verdict, and conditions are immediately clear.
Economic mechanism	20	The causal chain and opposing force are correct, intuitive, and linked to the chosen counterfactual.
Data, benchmarks, and exhibits	25	Primary data and informative comparisons are verified, reproducible, legible, and interpreted without overclaiming.
Tradeoffs and judgment	20	The strongest objection, distribution, uncertainty, sensitivity, and reversal condition genuinely test the verdict.
Four-page narrative	10	Concise complete prose, claim-based headings, and a coherent decision document rather than slide language.
Reproducibility and AI transparency	10	Clean files, working code, source audit, prompt record, corrections, and clear partner contributions.

Each dimension uses a 0–5 performance level: **5 excellent**, **4 strong**, **3 competent**, **2 partial**, **1 insufficient**, and **0 missing**. A high-scoring memo need not make a dramatic claim; its confidence must match its evidence.

29. Final upload checklist

- One PDF: four-page memo + at most one-page references/data appendix + at most one-page reproducibility/AI appendix.
- One ZIP: README, raw and processed data, code, generated figures, sources, verification report, AI log, and contribution statement.
- File names: `LastName-LastName-memo.pdf` and `LastName-LastName-research.zip`. One partner submits for the pair.
- Open the uploaded files from Canvas and confirm they are the final versions.

The final test. Could a skeptical reader reproduce your exhibits, understand the central tradeoff, see the strongest case against you, and know exactly when your recommendation should be reversed? If yes, the memo is doing its job.